

Aqara FP300 Installer Guide

1. Hardware Overview

The FP300 is a battery-powered multi-sensor combining:

- **60 GHz mmWave radar** — stationary presence detection
- **PIR sensor** — fast initial motion triggering
- **Temperature, humidity, and illuminance sensors**

It communicates over **Zigbee** (full feature set via Zigbee2MQTT) or Matter-over-Thread (limited parameter exposure). Always use **Zigbee mode** with Z2M for complete configuration control.

Detection specifications:

- Maximum detection range: **6 m**
- Field of view: **120°**
- Battery life: ~1 year (varies with polling intervals and room activity)

2. Physical Installation

2.1 Mounting Methods

Method	Max Height	Notes
Adhesive	2 m	Included in box; reversible
Magnetic	2 m	Easiest to reposition during testing
Expansion screws	No limit	Required above 2 m

Tip: Use the magnetic mount during initial setup and testing. Only commit to adhesive or screws once you have validated coverage.

2.2 Height Guidelines

Height	Requirement
1.4 1.8 m	Optimal for wall/corner mount
~2.0 m	Tilt sensor slightly downward toward activity zone
> 2.0 m	Screws mandatory; angle must compensate for height
Ceiling	Valid but reduces effective range; not recommended for rooms > 15 m

2.3 Placement Strategy

Corner placement is strongly preferred over flat-wall mounting. A corner position with the sensor facing the room center delivers the widest combined coverage and minimizes blind zones at room edges.

Avoid placing near:

- Air conditioning vents or outlets
- Ceiling fans or air purifiers
- Large metal or glass surfaces (reflections cause false detections)
- Positions where the detection cone crosses into adjacent rooms through walls

2.4 Near-Field Dead Zone

Detection below **1 m** from the sensor face is unreliable. Account for this when positioning — do not place the sensor where occupants regularly stand directly in front of it at close range.

2.5 Pre-Installation Recommendation

Before permanently fixing the sensor, use painter's tape or the magnetic mount to trial the position for **24–48 hours**. This reveals blind spots and interference sources before committing to a final installation.

3. Pairing with Zigbee2MQTT

1. In Z2M UI, click **Permit join**
2. Hold the sensor button for **5 seconds** until the LED flashes — this triggers pairing mode
3. The device will appear in Z2M as [Aqara PS-S04D](#)
4. Rename it in Z2M UI (e.g. [fp300_living_room](#))
5. Verify the **Exposes** tab shows all parameters: [presence](#), [temperature](#), [humidity](#), [illuminance](#), [detection_range](#), etc.

4. Configuration Correct Order

■■ **Always apply all settings before running Spatial Learning.** The sensor calibrates its empty-room baseline against the active configuration. Changing settings after learning degrades accuracy and requires a new learning session.

Important: Understanding Z2M UI Behavior

The FP300 is a battery-powered sleeping device. When you change a parameter value in the Z2M Exposes tab, the change is **immediately sent** to the device, but the sensor only processes it when it next wakes up — which can take **several minutes** depending on room activity.

To force immediate feedback after changing settings:

1. Walk up to the sensor
2. **Press the button once** (short press)
3. The device wakes immediately and processes all pending commands within a few seconds

The Z2M UI will update automatically once the sensor reports back. No "Apply" button is required — each toggle/dropdown sends its own command instantly.

Step 1 Set Detection Mode

Z2M Parameter	Recommended Value	Notes
presence_detection_options	mmwave	Eliminates PIR-induced false triggers
pir_detection_interval	60 sec	Only relevant if using both ; limits PIR re-trigger rate

Use **both** only if fast light-on response (< 1 sec) is a hard requirement. For all other cases, **mmwave** provides cleaner and more reliable presence data.

Step 2 Configure Sensitivity

Z2M Parameter	Value	Notes
motion_sensitivity	See room profiles below	Controls mmWave radar sensitivity (not PIR) core detection tuning
ai_sensitivity_adaptive	ON	Allows the sensor to self-tune over time

***Parameter naming clarification:** Despite the name **motion_sensitivity**, this parameter controls the **mmWave radar** sensitivity for stationary presence detection. It remains the primary tuning parameter even when **presence_detection_options** is set to **mmwave** only.*

Step 3 Set Absence Delay Timer

Z2M Parameter	Range	Notes
absence_delay_timer	10 300 sec	See room profiles below

This timer defines how long after the last detected movement the sensor waits before declaring the room empty. Setting it too low causes false absences when occupants are sitting still.

Step 4 Restrict Detection Range

All 24 **detection_range_X** bands (each representing **0.25 m**) default to **true**, meaning the sensor scans the full 0–6 m including through walls. Restrict to your actual room depth.

How to calculate:

- Divide your room's maximum occupied depth (in metres) by 0.25 → number of bands to enable
- Always disable **detection_range_0** through **detection_range_3** (covers 0–1 m dead zone near mounting wall)
- Disable all bands beyond your room boundary

Example — 4 m room, sensor in corner:

Bands	Range	Value
detection_range_0_3	0 1 m	false (dead zone)
detection_range_4_15	1 4 m	true
detection_range_16_23	4 6 m	false (beyond room)

Step 5 Enable AI Interference Detection

Z2M Parameter	Value	Notes
ai_interference_source_selfidentification	ON	Learns to ignore fans, AC drafts, convection

Step 6 Run Spatial Learning

This is always the **last step**, after all other settings are saved.

1. **Empty the room completely**
2. In Z2M UI → Exposes tab → set **spatial_learning** → **Start Learning**
3. Stay out for a minimum of **60 seconds**
4. No confirmation appears in the UI or log — this is expected behavior
5. The sensor continues autonomous background learning indefinitely after the initial trigger

If the room was occupied during any previous learning sessions, those baselines were contaminated. Each new session fully overwrites the previous one.

5. Room-Type Configuration Profiles

Living Room

Scenario: extended occupancy, often seated and still; TV, fans, or AC common.

Parameter	Value
presence_detection_options	mmwave
motion_sensitivity	medium
ai_sensitivity_adaptive	ON
ai_interference_source_selfidentification	ON
absence_delay_timer	90 sec
Detection range	Limit to seating area depth (typically 35 m)
Mount position	Corner, 1.41.8 m, aimed at sofa/seating area

Bedroom

Scenario: sleeping occupant with minimal movement; critical to avoid false absence during sleep.

Parameter	Value
presence_detection_options	mmwave
motion_sensitivity	high
ai_sensitivity_adaptive	ON
ai_interference_source_selfidentification	ON
absence_delay_timer	180 sec
Detection range	Limit to bed distance (typically 24 m)

Parameter	Value
Mount position	Corner, 1.41.7 m, aimed directly at bed

Note: The FP300 does **not** support sleep monitoring (breathing/micro-movement detection). That feature is exclusive to the Aqara FP2. High sensitivity enables detection of very still occupants but is not equivalent to sleep monitoring.

Bathroom / Toilet

Scenario: short visits; fast on/off required; humidity sensor useful for fan automation.

Parameter	Value
presence_detection_options	both
motion_sensitivity	low
ai_sensitivity_adaptive	ON
ai_interference_source_selfidentification	ON
absence_delay_timer	30 sec
Detection range	Tight limit to room depth (typically 12.5 m)
Mount position	High corner above door, angled toward room interior

Bonus: Use the built-in humidity sensor to trigger an extractor fan when **humidity > 70%**, independent of presence detection.

Hallway / Corridor

Scenario: brief transit only; high risk of through-wall detection into adjacent rooms.

Parameter	Value
presence_detection_options	both
motion_sensitivity	low
ai_sensitivity_adaptive	ON
ai_interference_source_selfidentification	OFF
absence_delay_timer	15 sec
Detection range	Match corridor length exactly; no extra margin
Mount position	End wall facing along corridor length, or high corner

Home Office / Study

Scenario: single occupant seated at desk for extended periods; light automation common.

Parameter	Value
presence_detection_options	mmwave
motion_sensitivity	medium

Parameter	Value
<code>ai_sensitivity_adaptive</code>	ON
<code>ai_interference_source_selfidentification</code>	ON
<code>absence_delay_timer</code>	60 sec
Detection range	Limit to desk distance (typically 1.53 m)
Mount position	Corner beside or above desk, aimed at workstation

6. Sensitivity Quick Reference

Room depth	Recommended sensitivity
< 4 m	low
4 – 7 m	medium
> 7 m	high

7. The Button

Action	Function
Single short press	Forces immediate wake-up; processes pending Z2M commands instantly
Hold for 5+ seconds	Factory reset / re-pairing (also switches Zigbee Thread)
Short press 10 times	Forces network rejoin

Tip: After changing any settings in Z2M, press the button once to force immediate feedback rather than waiting several minutes for the device to wake naturally.

8. Troubleshooting

Symptom	Likely Cause	Fix
Ghost detections at night	PIR reacting to thermal changes or drafts	Switch to <code>mmwave</code> only; enable <code>ai_interference_source_selfidentification</code>
False absences while sitting still	Sensitivity too low or timer too short	Increase <code>motion_sensitivity</code> ; raise <code>absence_delay_timer</code> to 60 sec
Through-wall detections	Detection range not restricted	Disable <code>detection_range</code> bands beyond room boundary
Persistent false detections after full setup	Spatial Learning run with room occupied	Re-run Spatial Learning with room completely empty
No presence detected close to sensor	Near-field dead zone (< 1 m)	Reposition sensor; keep <code>detection_range_0_3</code> disabled
Sensor reports absent immediately after leaving	<code>absence_delay_timer</code> too short	Increase to room-type appropriate value

Symptom	Likely Cause	Fix
Z2M settings not updating in UI	Device is asleep and hasn't processed command yet	Press button once to force immediate wake-up and processing